

AMERICAN TERA WATT

HVDC System Functional Specification

Private Industrial Power Grid

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Revision history

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1 Introduction

1.1 Purpose

This functional specification defines the functional and performance requirements for the HVDC transmission system to be supplied for the *Utah/Nevada Industrial Grid (UNIG)* scheme and is the basis for HVDC system supplier selection.

1.2 Scope

The scope of supply is the HVDC terminals, together with any additional equipment that the supplier considers necessary for the integration of the HVDC system into the grid.

The DC line route and the DC tower design are under development and are not fixed at the time of issue of this specification. The resulting DC line parameters will be provided as they become available; for the associated dates refer to Section 6.

1.3 Key project characteristics and design drivers

The following characteristics drive the design and execution of the scheme and should be considered by the supplier:

- **Islanded operation.** The system operates as an islanded network with *no interconnection to the wider AC grid*. The HVDC link and its connected generation and loads form an islanded network, which the converter control must be able to establish and stabilise.
- **Staged generation and transmission buildout.** Generation is built out in 300 MW steps, with the pace set by water availability. staged buildout of the transmission line, phased to match the generation ramp, is therefore beneficial and is a design objective.
- **High reliability requirements.** The data-center loads impose high reliability and availability requirements on the transmission system (see Section 4.1).
- **Standardized hardware design.** A standardized hardware design approach shall be adopted to reduce hardware lead times and to limit configuration variations between schemes.

2 System overview

2.1 Grid configuration

The HVDC system is a point-to-point, bidirectional VSC-HVDC link connecting the converter stations via a DC transmission line. An overview of the system with HVDC scope of supply is shown in Figure 1.

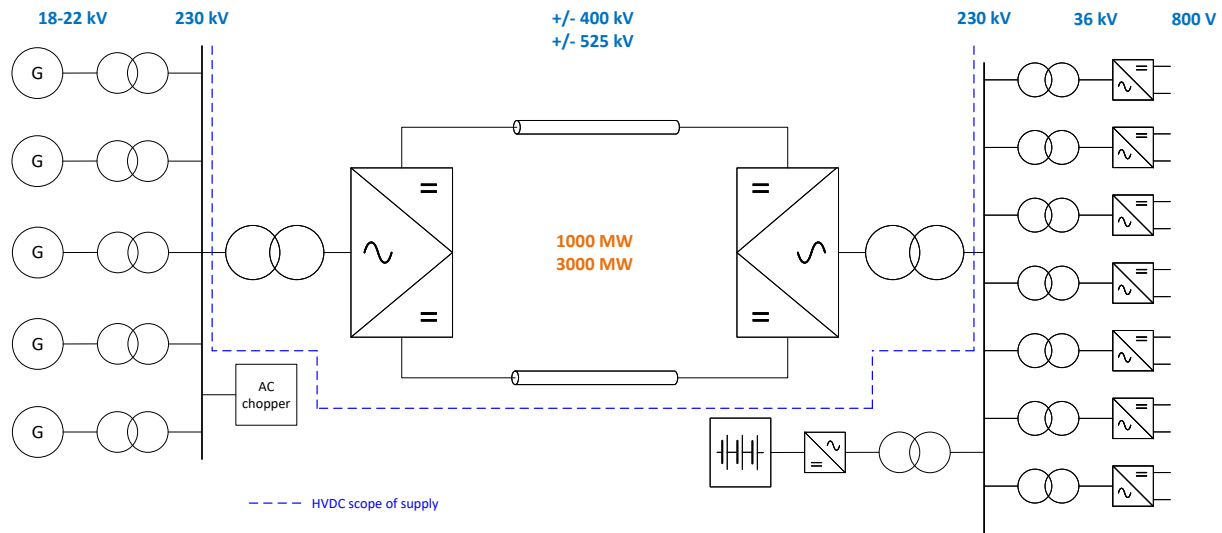


Figure 1: System overview with HVDC system scope of supply

2.2 Transmission schemes

Two HVDC schemes are currently under investigation. The schemes differ in their power rating, which results in different transmission topologies. An overview of the technical parameters is given in Table 1.

Parameter	Description	Scheme 1	Scheme 2
P_{AC_Rec}	Rated active power (AC rectifier)	1000 MW	3000 MW
Q_{AC_Rec}	Rated active power (AC rectifier)	± 330 MVar	± 990 MVar
U_{DC}	Rated DC voltage (pole–ground)	± 400 kV	± 400 kV, ± 525 kV
Config.	Transmission configuration	SMP	DSMP, BIP with DMR
L	DC line length	~ 200 mi	~ 200 mi
f	AC system frequency	60 Hz	60 Hz

Table 1: HVDC configuration: Scheme 1 and Scheme 2.

For the reactive power an initial power factor of $\cos \phi = 0.95$ was defined. As the system investigations are ongoing, it is expected that the reactive power requirements can be significantly reduced.

2.3 HVDC topologies

2.3.1 Scheme 1

The anticipated transmission configuration for Scheme 1 is a symmetrical monopole with a DC voltage rating of ± 400 kV (SMP, Figure 2). The 400 kV is considered to be the most economical solution for the transmission line considering converter and DC line costs. HVDC supplier is encouraged to alternative solutions in case of significant savings with smaller DC voltage set-up such as ± 320 kV.

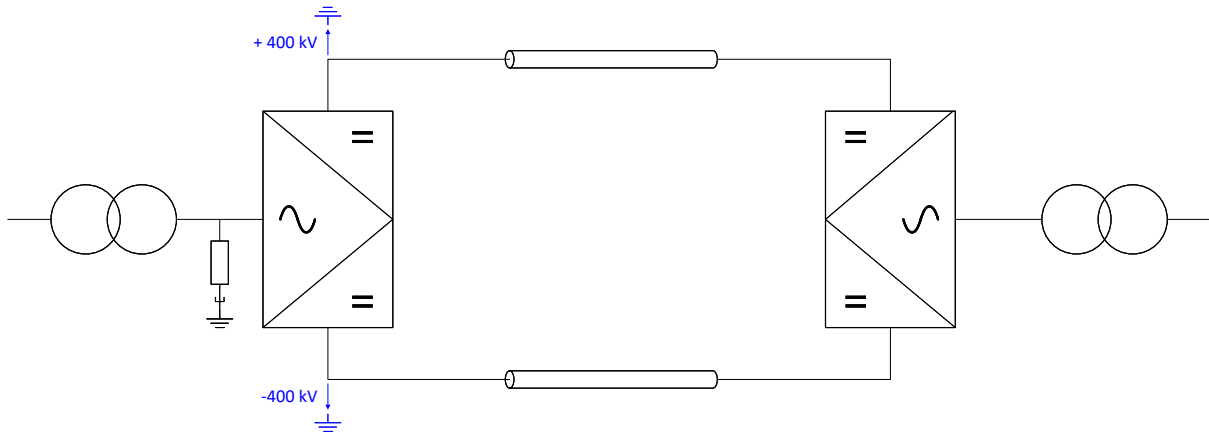


Figure 2: Symmetrical monopole (SMP) configuration.

2.3.2 Scheme 2

For the 3000 MW Scheme 2, two transmission configurations are considered: the dual symmetrical monopole (DSMP, Figure 3) and the bipole with dedicated metallic return (BIP, Figure 4). The DSMP configuration is assumed to provide higher availability and greater flexibility for the staged buildout of the transmission line. The decision between the DSMP and BIP configurations will be based on the price difference in the converter and DC line costs.

2.4 Generator

The generation connected to the HVDC system is provided by geothermal generators supplied by Turboden. Individual generator units are sized at app. 50 MW.

American Terawatt will provide a generator-system equivalent to be used for EMT-type studies and performance verification, together with an RSCAD model for hardware-in-the-loop (HIL) testing.

2.5 Load

The load connected to the HVDC system consists of data-center loads, which are characterised by power fluctuation.

American Terawatt will provide a data-center load equivalent to be used for EMT-type studies and performance verification, together with an RSCAD model for hardware-in-the-loop (HIL) testing.

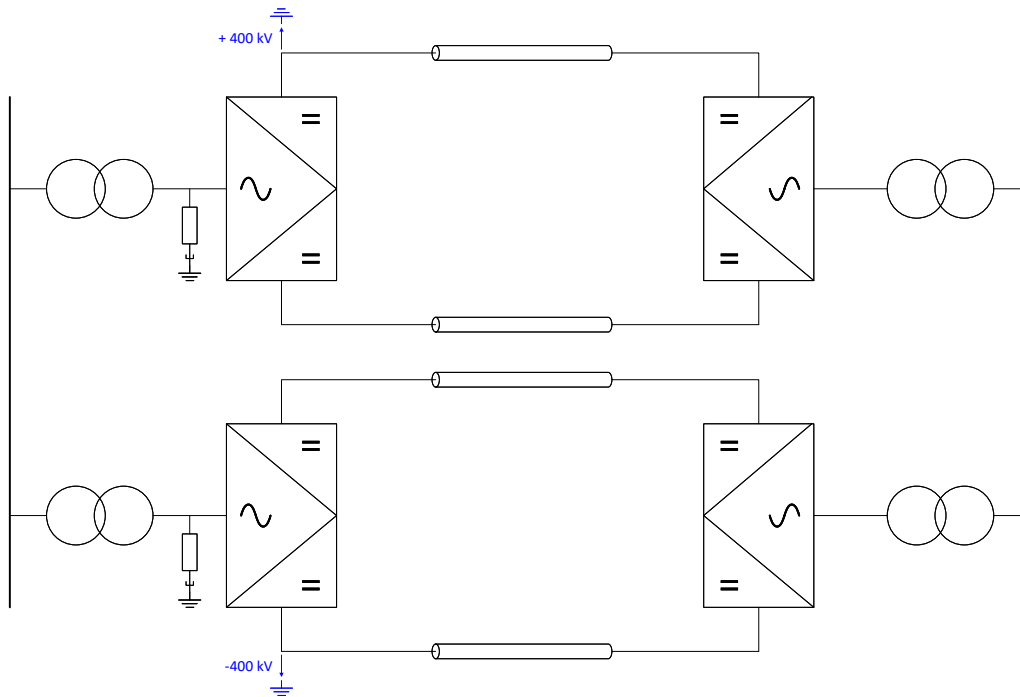


Figure 3: Dual symmetrical monopole (DSMP) configuration.

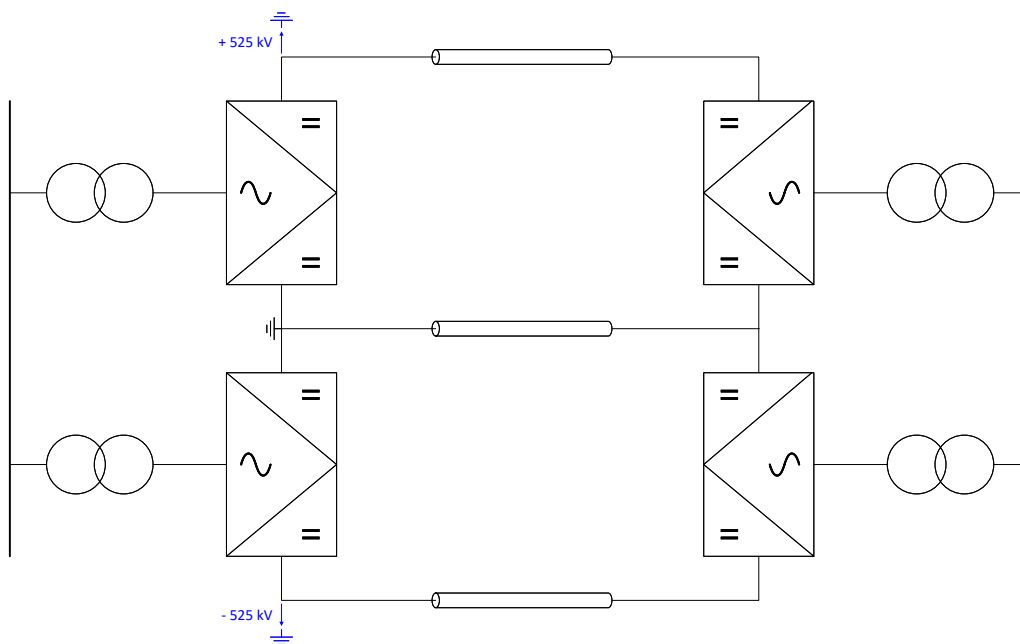


Figure 4: Bipole with dedicated metallic return (BIP) configuration.

3 AC data

The AC system data is flexible and is not fixed at the time of issue of this specification. A nominal AC voltage of 230 kV has been selected as a compromise between the insulation requirements, which grow with the system voltage (clearances, creepage, BIL and the resulting station and tower footprint), and the current requirements, which grow as the voltage is reduced (conductor and busbar sizing, switchgear and transformer current ratings, and losses). The supplier is encouraged to propose an alternative AC voltage where this results in a significant cost or performance benefit.

The interconnection requirements follow IEEE 2800. The steady-state parameters at the point of measurement (POM) are given in Table 3. The insulation coordination for the 230 kV system follows IEEE 1313.1 and IEEE C57.12.00 and is given in Table 4.

Unit		Value
Nominal system voltage	kV	230
U_m	Maximum system voltage (insulation class)	242 kV
COG	Coefficient of earthing (effectively earthed system)	≤ 1.4

Table 2: AC voltage

Concept verification studies are ongoing. The operating range values given in Table 3 and Table 4 are preliminary and will be adjusted based on the outcome of this investigation.

Parameter	Description	Value
U_n	Nominal AC system voltage (phase–phase, rms)	230 kV
U_m	Maximum continuous system voltage	242 kV
f_n	Nominal AC system frequency	60 Hz
—	Number of phases / phase sequence	3, ABC
—	Neutral earthing	Effectively earthed ($X_0/X_1 \leq 3$, $R_0/X_1 \leq 1$)
ΔU	Continuous operating voltage range at POM	0.90–1.05 pu
Δf	Continuous operating frequency range at POM	58.8–61.2 Hz
df/dt	Rate-of-change-of-frequency withstand	± 5 Hz/s (measured over 0.1 s)
Q	Reactive capability at rated P	± 0.329 pu (0.95 lead/lag)
u_2	Continuous negative-sequence voltage unbalance	≤ 1 %
THD_u	Voltage total harmonic distortion (IEEE 519, >161 kV)	≤ 1.5 % (individual ≤ 1.0 %)
P_{st}	Short-term flicker severity (IEEE 1453)	≤ 0.35
I_{sc}	Short-circuit level at POM	[TBC]

Table 3: AC steady-state parameters at the POM according to IEEE 2800 (preliminary).

Parameter	Description	Value
U_n	Nominal system voltage	230 kV
U_m	Maximum system voltage (insulation class)	242 kV
COG	Coefficient of earthing (effectively earthed system)	≤ 1.4
BIL	Lightning impulse withstand, 1.2/50 μ s	900 kV peak
—	Chopped-wave withstand ($1.1 \times \text{BIL}$)	990 kV peak
BSL	Switching impulse withstand, 250/2500 μ s	745 kV peak
—	Power-frequency short-duration withstand, 1 min dry	395 kV rms
U_c	Surge arrester MCOV ($\geq U_m/\sqrt{3}$)	144 kV rms
U_r	Surge arrester duty-cycle rating	180 kV rms
—	Minimum specific creepage distance	25 mm/kV ($[TBC]$, per site pollution class)

Table 4: Insulation coordination for the 230 kV AC system according to IEEE 1313.1 and IEEE C57.12.00 (preliminary).

4 Performance requirements

4.1 Availability

4.1.1 Data-center availability requirement

The transmission system supplies data-center loads, whose availability is specified against the Uptime Institute tier framework. It is assumed that a **Tier III** requirement applies. Tier III describes a *concurrently maintainable* facility: every element of the power (and cooling) path can be maintained or replaced without interrupting supply to the load, achieved through redundant capacity components and multiple distribution paths (with one path active at a time). In availability terms, Tier III corresponds to about **99.982 %** availability — roughly **1.6 hours** of downtime per year.

A single HVDC link cannot, on its own, meet this target; achieving it requires redundancy at system level (e.g. dual full-capacity circuits) together with battery storage and, where necessary, on-site generation. The availability of the HVDC system is one contributor to the overall facility availability and is specified below.

4.1.2 Guaranteed values

Operational performance shall be measured per CIGRE TB 956. Guaranteed values are given in Table 5.

Metric	Description	Guaranteed
FEU	Forced energy unavailability	$\leq 0.5\%$
SEU	Scheduled energy unavailability	$\leq 1.0\%$
FOR	Forced outage rate SMP	$\leq 2 / \text{yr}$
FOR	Forced outage rate BIP (single pole)	$\leq 2 / \text{yr}$
FOR	Forced outage rate BIP (bipole)	$\leq 0.1 / \text{yr}$
EA	Energy availability	$\geq 98.5\%$

Table 5: Guaranteed availability and performance values.

Monitoring period should be 3 years, extendable to 4 with best 3 out of 4 years.

5 Environmental and site conditions

Station location is in Utah and Nevada ...

Parameter	Value
Ambient temperature range (T_{amb})	−15 ... +45 °C
Site altitude	1500 m
Design seismic level	IEEE 693-2018: Medium

Table 6: Environmental and site design conditions.

6 Time Line

The following milestones apply to the project. Dates marked *[TBC]* are indicative and will be confirmed as the design develops. The milestones are listed in Table 7.

Ref.	Milestone	Date
M1	DC line environmental analysis and DC line design	Nov. 2026
M2	HVDC specification	Dec. 2026
M3	NTP Link 1	Jun. 2027
M4	NTP Link 2	Jun. 2027
M5	NTP Link 3	Jun. 2027
M6	NTP Link 4	Jun. 2027
M7	COD Link 1	Jun. 2030
M8	COD Link 2	Jun. 2030
M9	COD Link 3	Jun. 2030
M10	COD Link 4	Jun. 2030

Table 7: Project timeline milestones.

References

1. ...
2. ...